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Impact of variable cosmology on the hydrogen 21cm cosmic dawn signal

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21cm cosmology

- Photons of 21cm wavelength are spontaneously produced by spin-flip transition in atomic hydrogen
- Enables the probing of events and processes during the Dark Ages, Cosmic Dawn, and the Epoch of Reionization

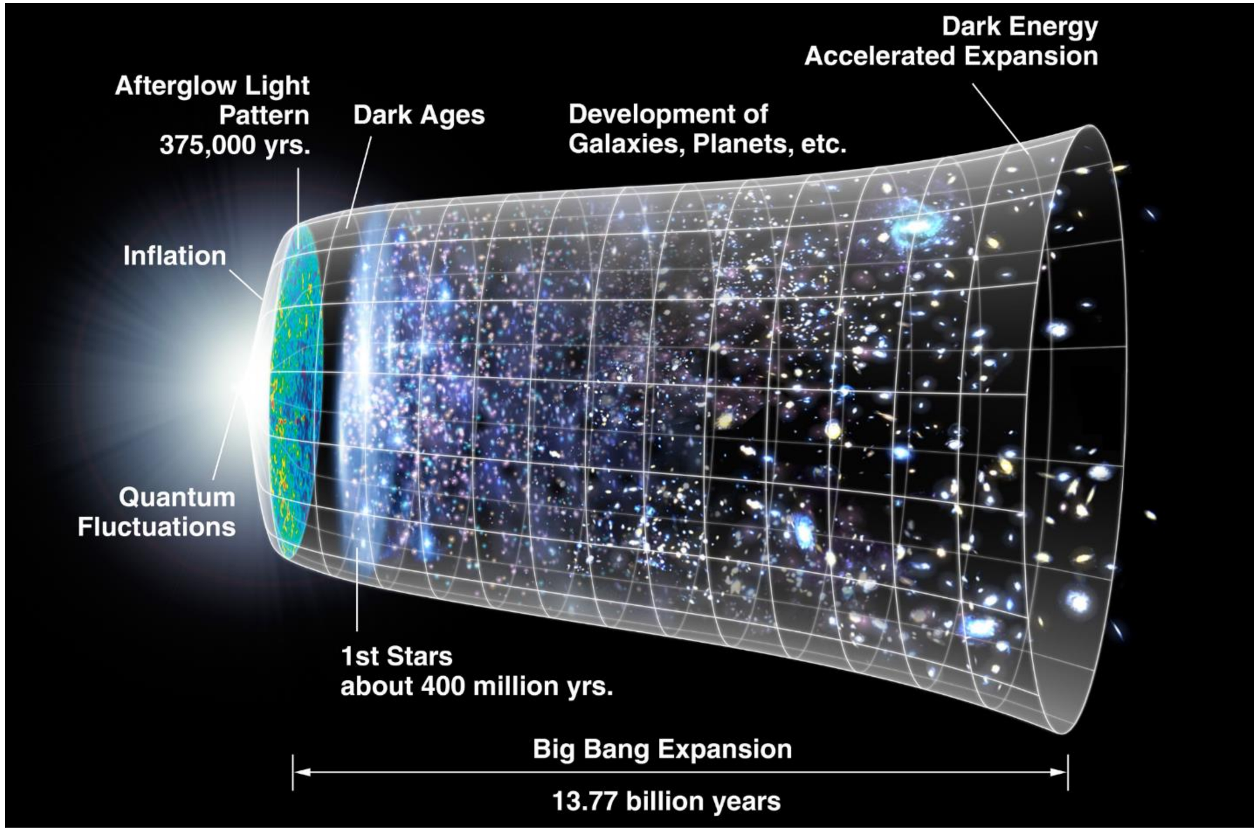


Image credit: NASA/WMAP Science Team

21cmSPACE

- 21cm Semi-numerical Predictions Across Cosmic Epochs (21cmSPACE)
- Outputs the 21cm global signal and 21cm power spectrum

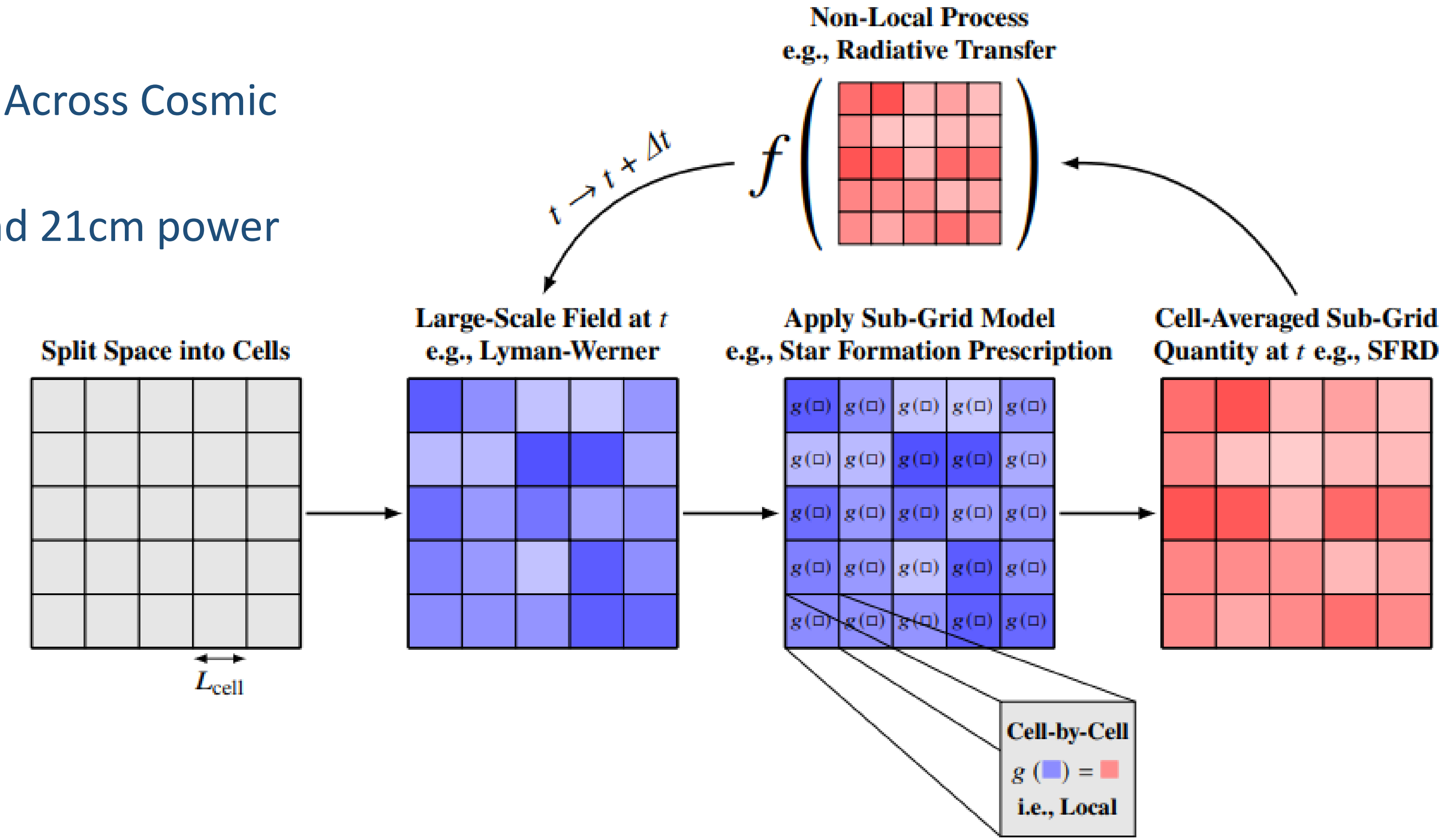


Image credit: Gessey-Jones, 2024

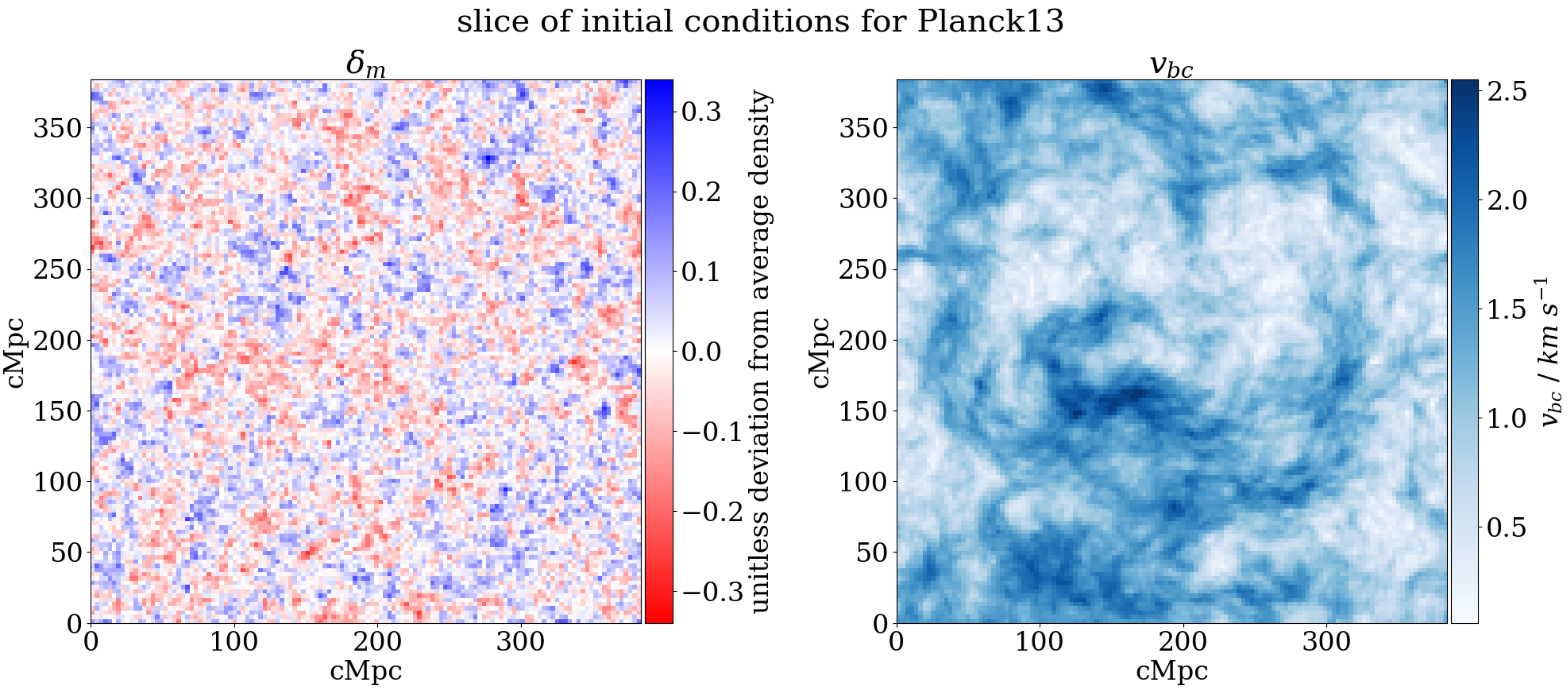
Time, space, cosmology

- Time
  - Begins at $z = 50$, and ends at $z = 6$
  - Timestep is 1 until $z = 15$, then 0.1 until $z = 6$
- Space
  - Total box size is variable by changing the number of pixels
    - 128^3, 256^3, or 512^3 pixels: 384^3 cMpc^3, 768^3 cMpc^3, or 1536^3 cMpc^3
    - Pixel size is hard-coded to 3 cMpc
- Cosmology
  - Hard coded to Planck 2013 ΛCDM best-fit model

Project aims

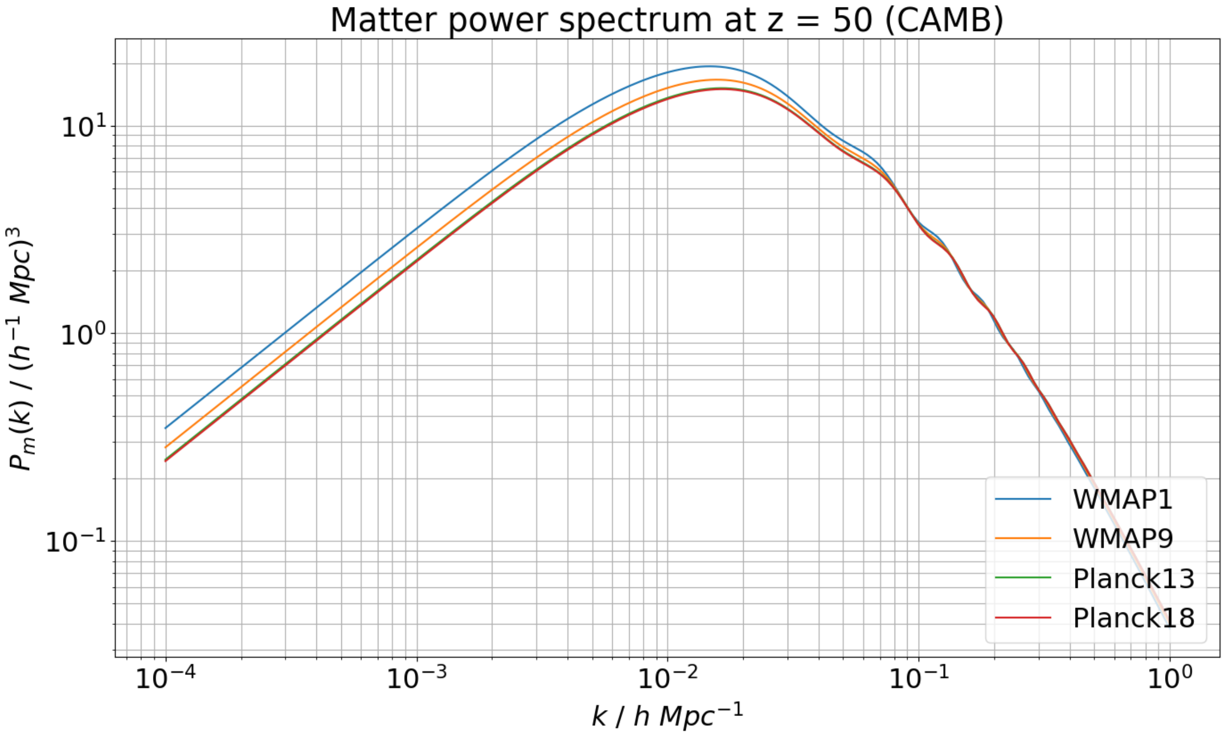
- Incorporate variable cosmology into 21cmSPACE
    - Remove hard coding
    - Enable exploration of cosmological parameter space
    - Hubble parameter, densities, CMB temperature
    - Initial conditions: redshift 50
- Allow simulation of signal for comparison with data
    - Verify or reject theories
    - Independent cosmological probe

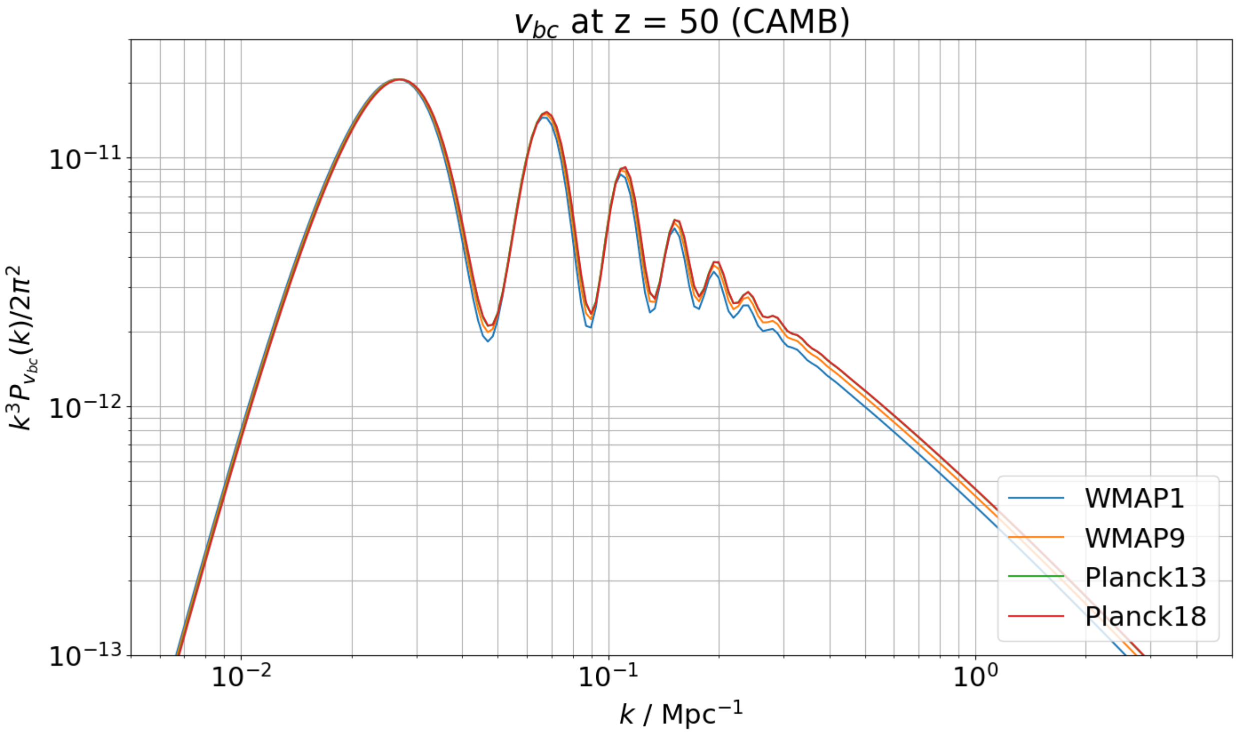
Initial condition grids (z = 50)



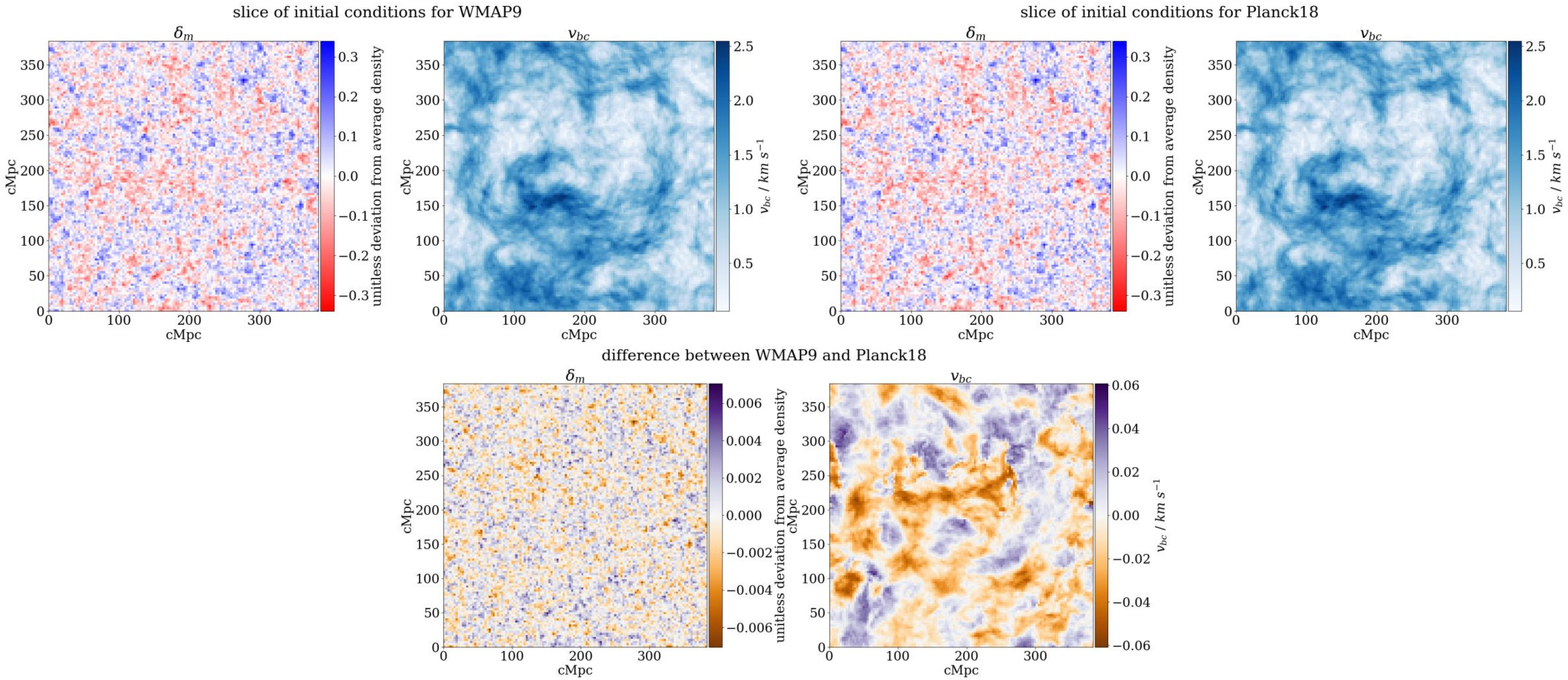
Initial conditions: matter power spectrum (z = 50)

$$k \in \left[\frac{2\pi}{L_{box}}, \frac{\pi\sqrt{3}}{L_{pix}}\right];\ \frac{2\pi}{L_{box}} \text{ is order } 10^{-2}\ \mathrm{Mpc}^{-1}, \frac{\pi\sqrt{3}}{L_{pix}} \text{ is order } 1\ \mathrm{Mpc}^{-1}$$





Initial condition grids: changing cosmologies (z = 50)



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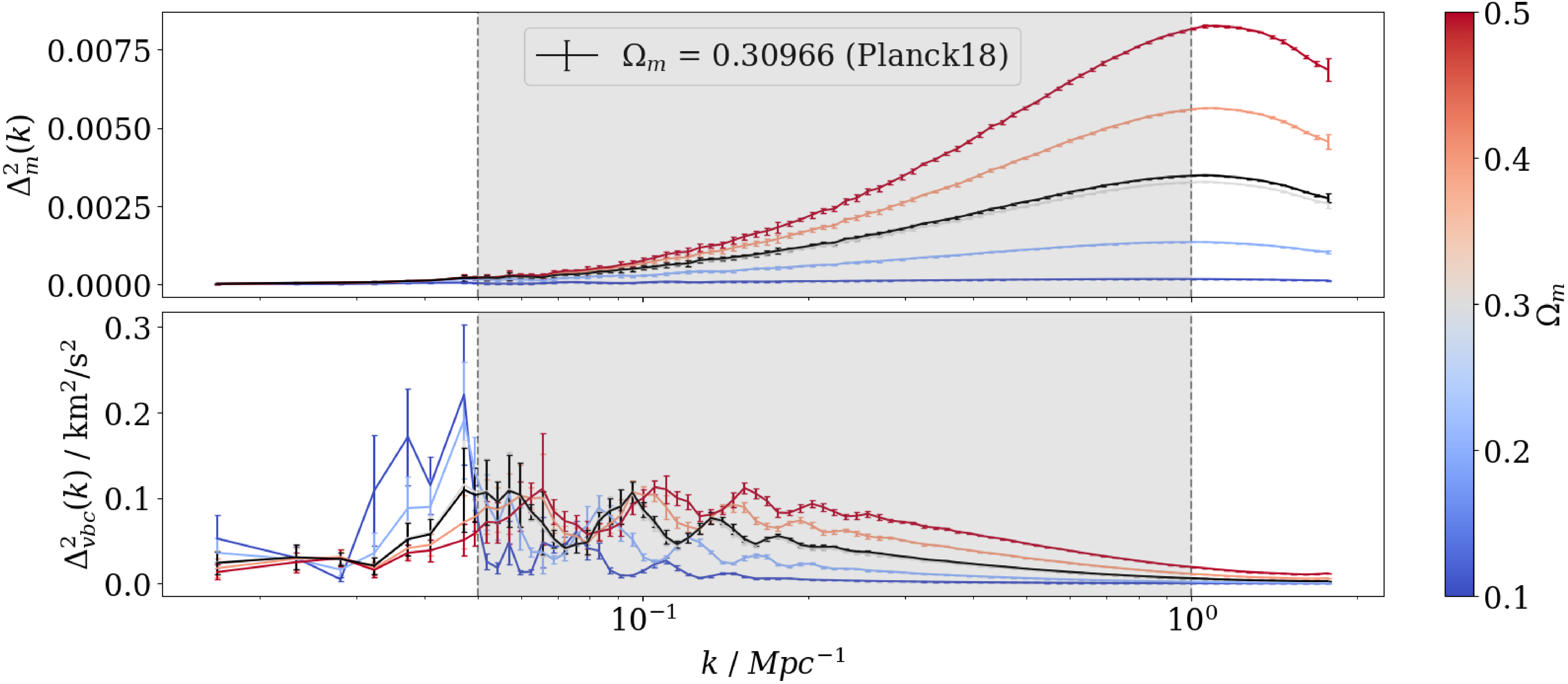
Custom cosmologies: parameter values

Cosmology	h	Ω_m	Ω_b	Ω_{dm}	Ω_{de}	T_{CMB} / K
Planck18	0.6766	0.30966	0.04897	0.26069	0.68885	2.7255
Om1	0.6766	0.1	0.04897	0.05103	0.89991	2.7255
Om2	0.6766	0.2	0.04897	0.15103	0.79991	2.7255
Om3	0.6766	0.3	0.04897	0.25103	0.69991	2.7255
Om4	0.6766	0.4	0.04897	0.35103	0.59991	2.7255
Om5	0.6766	0.5	0.04897	0.45103	0.49991	2.7255
Ob3	0.6766	0.30966	0.03	0.27966	0.69025	2.7255
Ob4	0.6766	0.30966	0.04	0.26966	0.69025	2.7255
Ob5	0.6766	0.30966	0.05	0.25966	0.69025	2.7255
Ob6	0.6766	0.30966	0.06	0.24966	0.69025	2.7255
Ob7	0.6766	0.30966	0.07	0.23966	0.69025	2.7255
h5	0.5	0.30966	0.04897	0.26069	0.69017	2.7255
h6	0.6	0.30966	0.04897	0.26069	0.69022	2.7255
h7	0.7	0.30966	0.04897	0.26069	0.69025	2.7255
h8	0.8	0.30966	0.04897	0.26069	0.69027	2.7255
h9	0.9	0.30966	0.04897	0.26069	0.69029	2.7255

Initial conditions: power spectra (z = 50)



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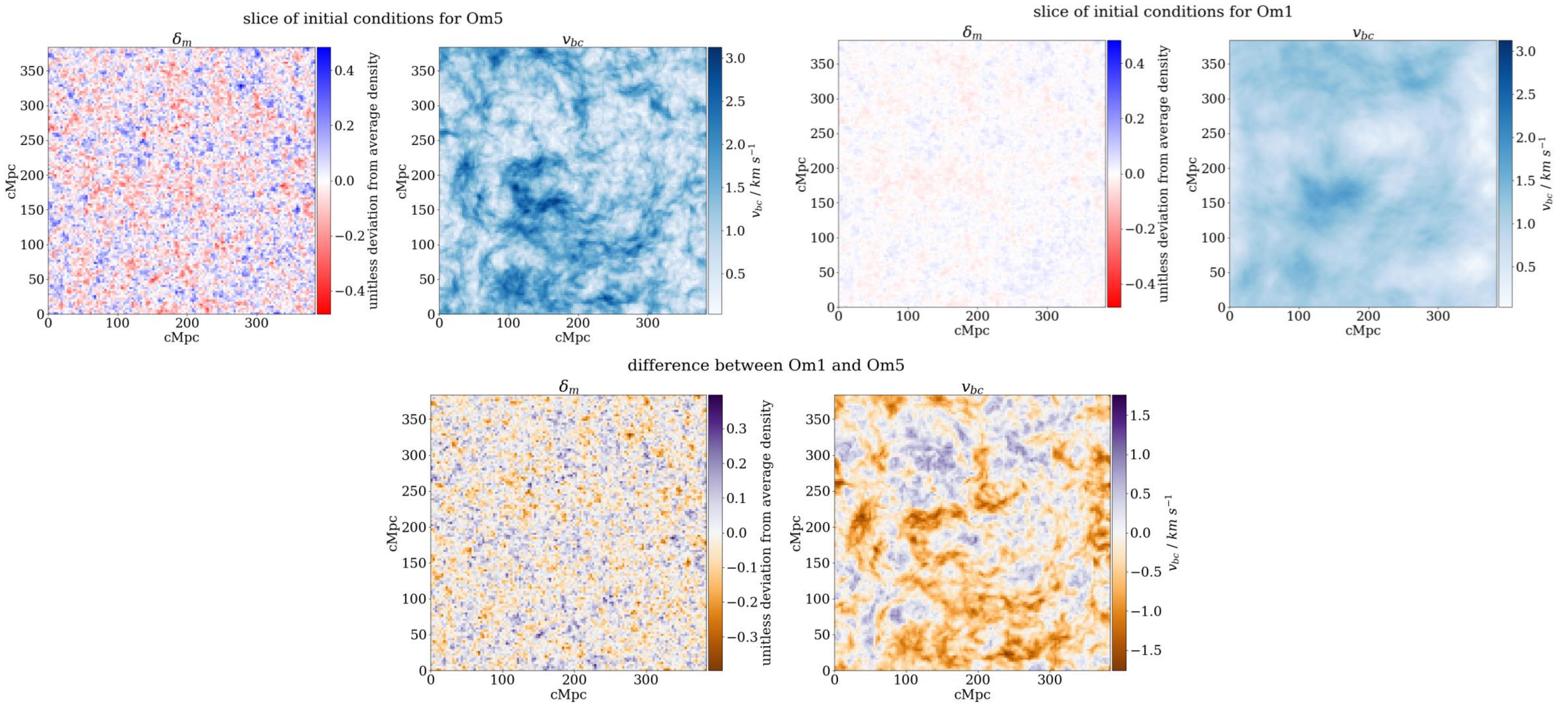
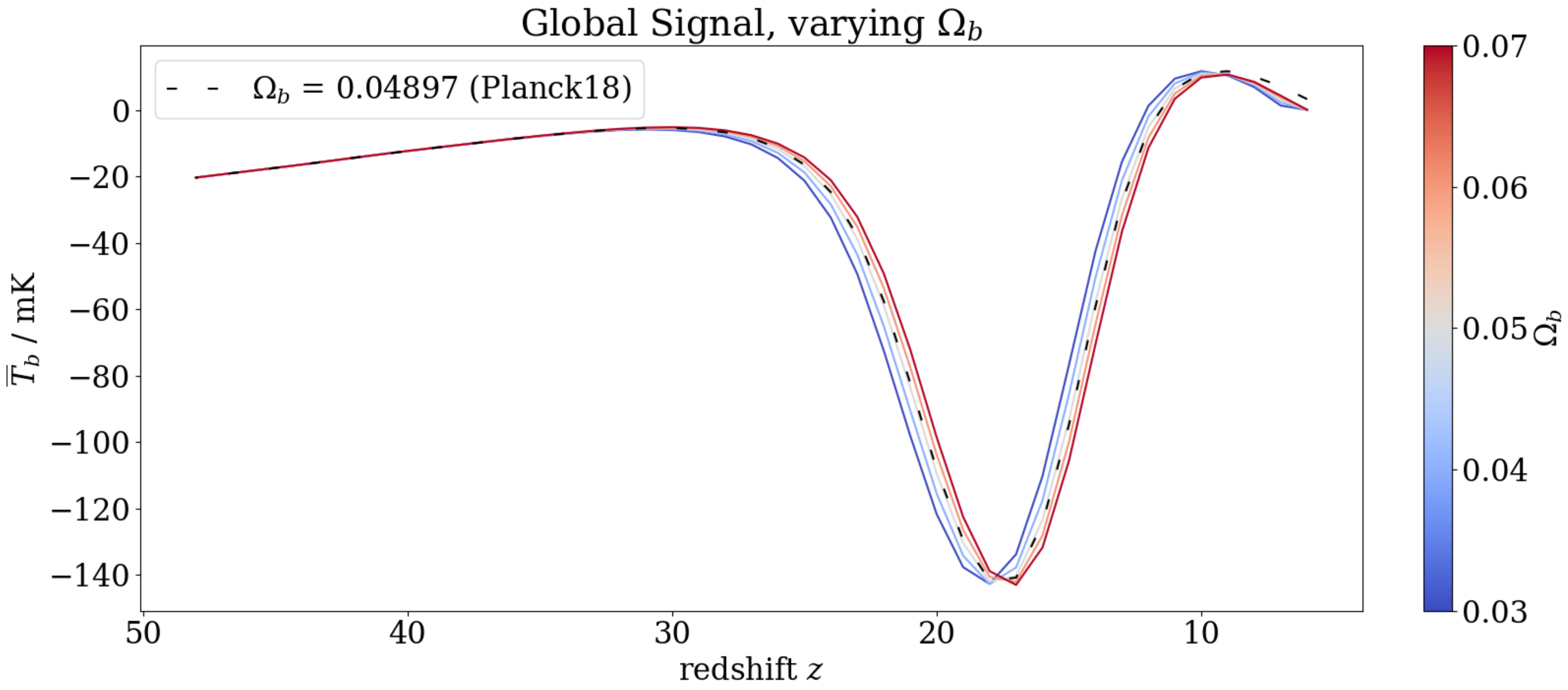
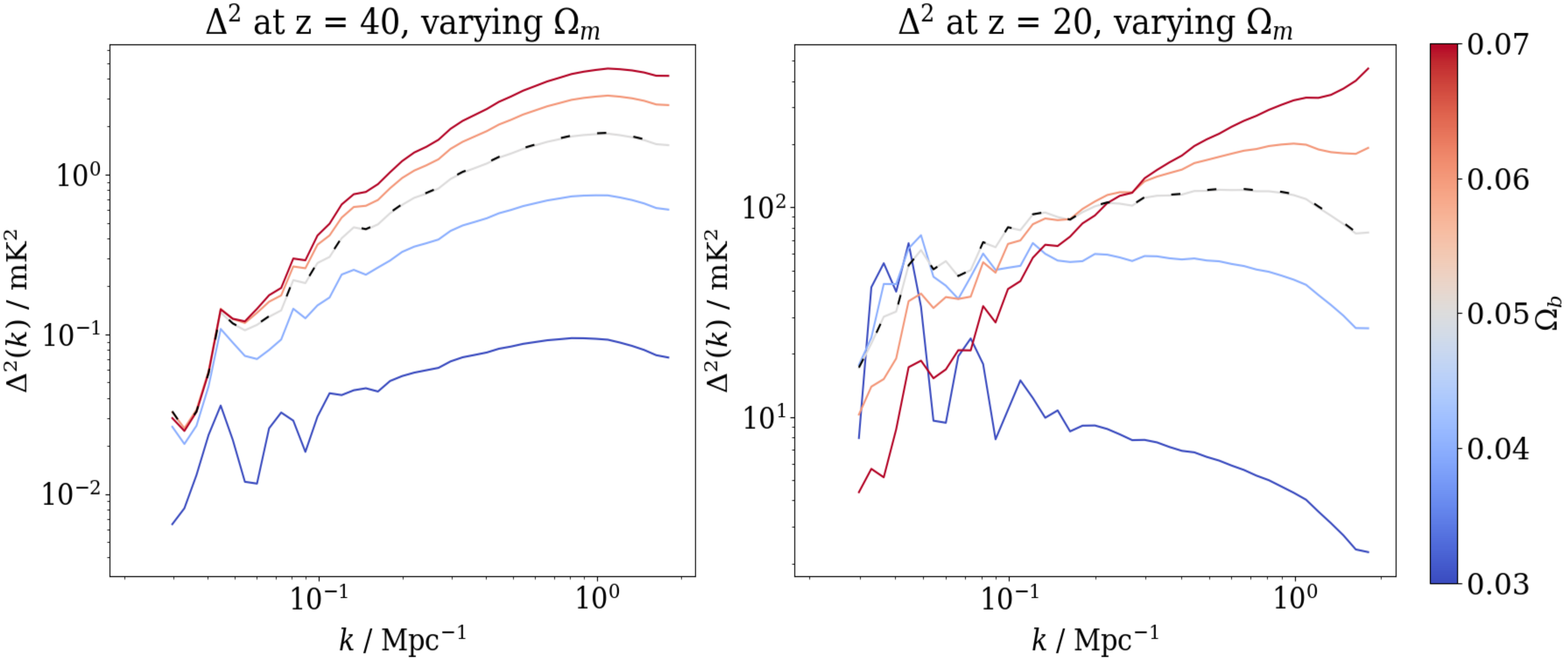
Initial condition grids: custom cosmologies (z = 50)



Results: varying Ω_m



Results: varying Ω_m



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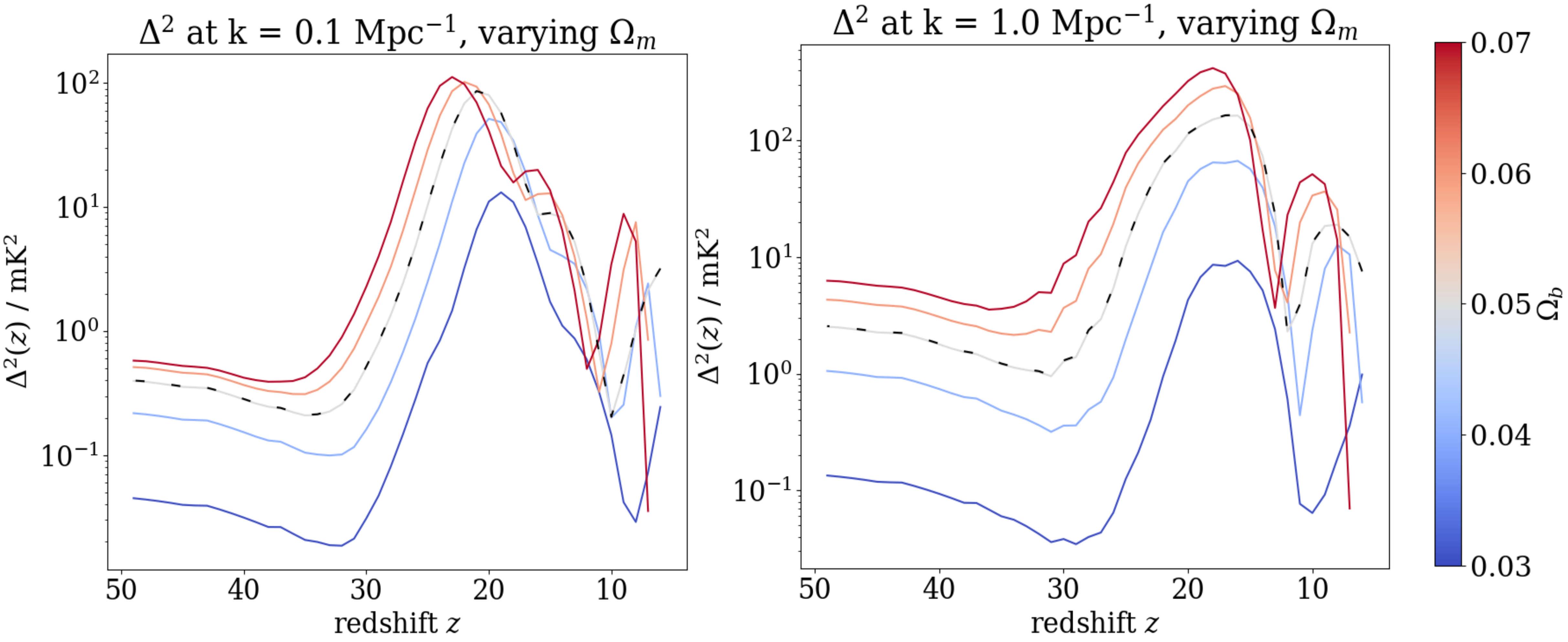
Results: varying Ω_m



Conclusions and future work

- Conclusions:
  - Infrastructure for cosmological dependence in initial conditions
  - Effect significantly propagated through 21-cm signal

- Future work:
  - Propagate cosmological parameters through evolution
  - Extend to Dark Ages
  - Improve parameter estimation pipelines